Principal Component Analysis (PCA)

Interpretation, Validation, Outliers, and Process Monitoring

Question Set with Solutions

TTK4260: Multivariate Data Analysis and ML

A. Interpretation: Scores, Loadings, Correlation Loadings, Biplots

Question 1 *(Score plot: clusters, gradients, outliers)*

[6 points] You run PCA with two components and plot a score plot (t_1 vs. t_2). You observe: (i) two tight clusters separated mainly along t_1, (ii) points within each cluster are smoothly ordered from left to right when colored by a continuous variable "feed rate", and (iii) one isolated point far in the top-right.

(a) What do the two clusters suggest?

(b) What does the smooth ordering with feed rate suggest (often called a "trend" or "gradient")?

(c) List two plausible reasons for the isolated point, and what you would check next.

Solution.

(a) Two tight, separated clusters typically indicate *two regimes/groups* in the data (e.g., two operating conditions, two product grades, two seasons, two populations). Separation along t_1 means PC1 captures the dominant difference between the groups.

(b) A smooth ordering (color gradient) implies a *continuous latent factor* is being captured by the scores. Here, feed rate is likely correlated with the direction of increasing t_1 (or some combination of t_1, t_2), so moving along that direction corresponds to increasing/decreasing feed rate or a process effect driven by it.

(c) The isolated point could be (1) a true abnormal event/fault (sensor failure, process upset), or (2) a data issue (unit mismatch, wrong scaling, missing-value imputation artifact, mislabeled sample). Next checks: inspect raw measurements for that sample; compute T^2 and Q; see whether it is high-T^2 (extreme but in-model) or high-Q (not explained by the model); verify preprocessing was identical; confirm metadata/labels.

Question 2 *(Score line plot: drift vs step vs cycles)*

[6 points] You plot the first score over time, $t_1(t)$ (a score line plot). Describe what each pattern suggests:

(a) a slow monotone increase in $t_1(t)$ over weeks,

(b) a sudden step change in $t_1(t)$ at one time,

(c) a periodic oscillation in $t_1(t)$,

(d) occasional isolated spikes.

For each case, give one typical process/measurement cause.

Solution.

(a) Slow monotone change suggests *drift* (gradual process change): catalyst aging, fouling, seasonal ambient effects, calibration drift.

(b) Step change suggests *an event* or operating-point switch: setpoint change, recipe change, maintenance intervention, sensor recalibration.

(c) Periodic oscillation suggests *cyclic dynamics*: daily load cycles, control-loop oscillation, batch cycle, rotating equipment effects.

(d) Isolated spikes suggest *transients/outliers*: brief disturbances, communication dropouts, sensor glitches, or short-lived process upsets.

Question 3 *(Loading plot: variable relationships)*

[7 points] In a loading plot (p_1 vs. p_2), variables A and B are close together; variable C is roughly opposite them (about 180°); variable D is about 90° away from A.

(a) What do these angles imply about pairwise correlations between (A, B), (A, C), and (A, D)?

(b) If A and B are far from the origin while D is near the origin, what does that say about their representation in PC1–PC2?

Solution.

(a) In a loading/correlation geometry, small angle $\Rightarrow$ positive correlation, 180° $\Rightarrow$ negative correlation, 90° $\Rightarrow$ near zero correlation. Thus: A and B are positively correlated; A and C are negatively correlated; A and D are approximately uncorrelated.

(b) Far from origin means the variable has large loadings and is strongly associated with the first two PCs (well represented). Near origin means PC1–PC2 explain little of that variable; it may require more PCs or it may be mostly noise/unique variance.

Question 4 *(Why correlation loading plots are useful even if you have loading plots)*

[8 points] You already have a loading plot (p_1 vs. p_2). Explain *two* reasons why a correlation loading plot can still be useful. Also explain what the 50% and 100% circles mean.

Solution. Two key reasons:

- **Comparable effect size:** Correlation loadings are bounded in $[-1, 1]$ and can be read directly as correlations $\tilde{p}_{jk} = \mathrm{corr}(x_j, t_k)$. Raw loadings p_{jk} depend on scaling/units and are not correlations.

- **Quality of representation:** The radius gives how well a variable is explained by the plotted PCs. For PC1–PC2,

$$r_j^2 = \mathrm{corr}(x_j, t_1)^2 + \mathrm{corr}(x_j, t_2)^2,$$

so distance from the origin indicates how much of variable x_j is captured by the 2D PCA model.

The **100% circle** (radius 1) corresponds to $r^2 = 1$ (variable fully explained by the plotted PCs). The **50% circle** corresponds to $r^2 = 0.5$, i.e. the plotted PCs explain 50% of that variable's variance (often drawn with radius $\sqrt{0.5} \approx 0.707$).

Question 5 *(Interpreting a correlation loading plot)*

[7 points] On a correlation loading plot for PC1–PC2, variable x_5 lies inside the 50% circle near the origin. Variable x_1 lies close to the outer circle in the positive t_1 direction.

(a) What does this imply about how well x_5 is modeled by the first two PCs?

(b) What does the position of x_1 imply about its relationship with t_1 and how it contributes to PC1 interpretation?

(c) Give one action you might take if several important variables lie inside the 50% circle.

Solution.

(a) Inside the 50% circle means $r^2 < 0.5$: PC1–PC2 explain less than half of the variance of x_5. It is poorly represented in the 2D model.

(b) Close to the outer circle and aligned with positive t_1 means $\mathrm{corr}(x_1, t_1)$ is large and positive (near $+1$) and $\mathrm{corr}(x_1, t_2)$ is small. Thus x_1 strongly defines PC1; samples with high t_1 tend to have high x_1.

(c) Likely add more PCs (increase r) or revisit preprocessing (e.g. scaling) so that the model captures those variables. Alternatively, if the variables are noise/irrelevant, consider excluding them.

Question 6 *(Biplot reasoning: samples relative to variable arrows)*

[6 points] In a biplot (scores + loadings), a sample point lies far in the direction of the arrow for variable "Temperature", and roughly opposite to the arrow for "Yield". What does this say about that sample's Temperature and Yield values relative to the dataset mean?

Solution. In a biplot, a sample projected along a variable arrow indicates a higher-than-average value for that variable (after preprocessing). Thus, being far in the direction of the Temperature arrow means the sample has *high Temperature*. Being opposite to the Yield arrow means the sample has *low Yield* relative to the mean. This is a fast way to read "which variables are high/low for a sample" in the reduced space.

B. Outlier Detection and Diagnostics (T^2, Q, Influence)

Question 7 *(T^2 vs Q quadrant interpretation)*

[10 points] Consider a T^2 vs Q plot with critical limits T^2_{crit} and Q_{crit} (forming four quadrants). For each quadrant, describe what it typically means and one recommended action:

(a) low T^2, low Q

(b) high T^2, low Q

(c) low T^2, high Q

(d) high T^2, high Q

Solution.

(a) **Low T^2, low Q (Normal):** typical sample, fits the PCA model well. Action: no action.

(b) **High T^2, low Q (High leverage / extreme but modeled):** unusual combination of *modeled* patterns; the sample is far in score space but still lies near the PCA subspace. Action: check influence on the model; verify if it is a legitimate operating point; consider robust PCA if many such points dominate.

(c) **Low T^2, high Q (New type / not modeled):** sample is not extreme in score space but has large residual; behavior not captured by retained PCs. Action: investigate unmodeled dynamics (need more PCs? nonlinearity? sensor fault?); examine Q contributions.

(d) **High T^2, high Q (Outlier):** both extreme and poorly fitted; strong candidate for fault or bad data. Action: immediate investigation; check raw data, sensors, and root cause; do not blindly refit PCA including it.

Question 8 *(Influence plot vs T^2–Q plot)*

[7 points] Some texts use an "influence plot" as *leverage vs residual* while others use T^2 vs Q.

(a) What is common between these two plots?

(b) What is a key difference in emphasis?

(c) A point has high leverage (or high T^2) but low Q. Is it automatically a fault? Explain.

Solution.

(a) Both combine a **score-space influence measure** (leverage or T^2) with a **residual-space measure** (Q) to diagnose atypical samples.

(b) T^2 has a clear probabilistic interpretation (Mahalanobis distance in score space with an F-based limit under assumptions), while leverage is a hat-value style measure of influence (often tied to regression diagnostics). Both are related, but T^2 is more standard in MSPC.

(c) No. High leverage / high T^2 with low Q means the sample is extreme but still *consistent with the model structure*. It may be a legitimate operating condition or rare-but-normal behavior. You would investigate context and influence, but it is not automatically "bad".

C. Preprocessing: Mean Centering, Scaling, and Categorical Variables

Question 9 *(Why mean centering matters)*

[7 points] Explain what can go wrong if you apply PCA to data that is not mean-centered. In particular, what does PC1 tend to capture if the data cloud is far from the origin?

Solution. PCA seeks directions of maximum variance around the origin. If data are not mean-centered, the origin is not at the center of the cloud. Then PC1 can be dominated by the **mean offset** (the direction from the origin toward the mean) rather than the true directions of variability. Result: the first component may mostly describe "where the cloud sits" instead of "how it varies", misleading interpretation and variance explained. Mean centering moves the cloud to the origin so PCs describe *variation about the mean*.

Question 10 *(Why scaling/normalization matters)*

[8 points] You have variables measured in different units (e.g., Temperature in °C, Pressure in bar, Flow in L/min).

(a) What happens if you do PCA on unscaled variables?

(b) When is scaling to unit variance (standardization) recommended?

(c) Give one situation where you might *not* want to scale.

Solution.

(a) Variables with larger numerical variance/units dominate PCs. PCA may mostly reflect the highest-variance unit rather than meaningful structure.

(b) Standardization is recommended when variables have different units/scales and you want each variable to contribute comparably (common in chemistry/process monitoring and survey data).

(c) If units/variances are inherently meaningful (e.g., power signal-to-noise is tied to physical variance; or you care about preserving energy/variance structure), scaling may destroy the intended weighting. Another case: if variables are already commensurate and measured with similar noise levels.

Question 11 *(Can we apply PCA to categorical variables?)*

[10 points] You have a dataset with:

- continuous variables: temperature, pressure
- categorical variables: operator ID (A/B/C), product grade (1/2/3)

(a) Why is "plain PCA" on raw categorical labels not valid?

(b) Name two reasonable approaches for mixed data and state a key caveat for each.

(c) If you one-hot encode categories and run PCA, how do you interpret loadings for those dummy variables?

Solution.

(a) Categorical labels are not numeric measurements with meaningful distances (e.g., grade 3 is not necessarily "3x" grade 1). PCA assumes Euclidean geometry and covariance structure on numeric variables, so raw labels break the assumptions.

(b) Two approaches:

- **One-hot encoding + PCA:** converts categories to binary indicators. Caveat: introduces collinearity (dummy variables sum to 1 within a category set), and results depend on scaling/centering choices for binaries.
- **Methods designed for categorical/mixed data:** Multiple Correspondence Analysis (MCA) for purely categorical; Factor Analysis of Mixed Data (FAMD) for mixed. Caveat: different underlying metrics and interpretation rules than standard PCA; you must explain the chosen metric.

(c) Dummy-variable loadings indicate association with a category being present. A large positive loading for "Operator=B" on PC1 means samples with Operator B tend to score high on PC1 (relative to mean coding). Interpretation is relative and should be done alongside the continuous-variable loadings and score patterns.

D. Model Selection: Leave-One-Out and k-Fold Cross-Validation

Question 12 *(Leave-One-Out (LOO) cross-validation for PCA)*

[10 points] You want to choose the number of PCs r using leave-one-out cross-validation (LOO).

(a) Describe the LOO procedure for PCA in words.

(b) A common RMSECV definition is

$$\text{RMSECV}(r) = \sqrt{\frac{\sum_{i=1}^{m} \|\boldsymbol{x}_i - \hat{\boldsymbol{x}}_{i,-i}\|^2}{m \cdot n}}.$$

What are m and n here?

(c) Why can LOO be overly optimistic when samples are strongly correlated (e.g., time series)?

(d) If the RMSECV curve decreases until $r = 4$ and then is nearly flat, what would you choose and why?

Solution.

(a) For each sample i, fit a PCA model (with r PCs) on the remaining $m - 1$ samples, reconstruct/predict the left-out sample $\hat{\boldsymbol{x}}_{i,-i}$ using that model, compute reconstruction error, and average over all left-out samples. Repeat for each candidate r.

(b) m is the number of samples (observations/rows). n is the number of variables (features/columns).

(c) In correlated data, leaving out one sample still leaves many very similar samples in training (e.g., neighboring time points). The model "sees almost the same point" and error looks artificially small, so the CV error underestimates true generalization.

(d) Prefer $r = 4$ or possibly the *simplest model near the minimum* (e.g., $r = 3$–4) depending on how flat the curve is. The principle is to avoid unnecessary PCs that mostly fit noise while keeping enough to capture structure.

Question 13 *(k-fold cross-validation for PCA)*

[10 points] Compare k-fold CV to LOO for choosing the number of PCs.

(a) Describe k-fold CV for PCA.

(b) Give two advantages of k-fold over LOO in practice.

(c) What is "repeated k-fold" and why might it be used?

(d) For time series, what fold design is safer than random folds?

Solution.

(a) Split the dataset into k folds. For each fold f, fit PCA (with r PCs) on the other $k - 1$ folds and compute reconstruction error on the held-out fold. Average the error across folds; repeat for candidate r.

(b) Advantages: (i) much less computational cost than LOO when m is large (only k fits per r instead of m), and (ii) often less optimistic than LOO, producing more realistic error curves (especially when using appropriate fold structure).

(c) Repeated k-fold repeats the random partitioning multiple times and averages the RMSECV. This reduces variance due to a single lucky/unlucky split and yields a more stable model selection.

(d) Use **blocked** or **forward-chaining** folds that respect temporal order (do not mix future samples into training for past validation). This prevents leakage due to autocorrelation.

E. Computing PCs: Algorithm Choice

Question 14 *(Choosing a PCA algorithm)*

[12 points] Match each scenario to a suitable PCA computation method and justify briefly:

(a) Small n (e.g., $n = 20$), moderate m (e.g., $m = 500$), no missing values.

(b) Large m (millions), moderate n (hundreds), need the first 5 PCs.

(c) Very large sparse matrix, need approximate first 10 PCs quickly.

(d) Data contains missing values you do not want to impute upfront.

Choose from: eigen-decomposition of covariance, SVD of $\mathbf{X}$, randomized SVD, NIPALS.

Solution.

(a) **Eigen-decomposition of covariance** (or standard SVD) works well: n is small so forming and decomposing an $n \times n$ covariance matrix is cheap.

(b) **SVD of X** (possibly incremental/streaming variants) is robust; with large m you typically avoid explicitly forming a huge covariance in memory. If only a few PCs are needed, randomized/incremental SVD is also attractive.

(c) **Randomized SVD**: designed to approximate leading PCs efficiently for large-scale problems with limited memory/time.

(d) **NIPALS**: can handle missing data naturally by iterating and updating scores/loadings without requiring full imputation (commonly used when only a few PCs are needed).

F. PCA for Process Monitoring (MSPC): From Insight to Action

Question 15 *(Building an MSPC model with PCA)*

[14 points] Outline a practical workflow to use PCA for process monitoring. Your answer should mention: (i) preprocessing, (ii) model fitting on baseline data, (iii) how to compute monitoring statistics (T^2 and Q), (iv) how to set control limits, (v) what you do after an alarm.

Solution. A typical MSPC workflow:

(i) **Preprocess:** collect an in-control baseline dataset; handle missing data; mean-center; often scale to unit variance; optionally filter obvious gross errors.

(ii) **Fit PCA:** choose number of PCs r (scree + explained variance + CV); fit PCA to baseline to get loadings $\mathbf{P}$, scores $\mathbf{T}$, eigenvalues λ_k.

(iii) **Online monitoring:** for each new sample $\boldsymbol{x}(t)$, apply the same preprocessing, compute scores $\boldsymbol{t}(t) = \boldsymbol{x}(t)\mathbf{P}_r$, reconstruct $\hat{\boldsymbol{x}}(t) = \boldsymbol{t}(t)\mathbf{P}_r^\top$, residual $\boldsymbol{e}(t) = \boldsymbol{x}(t) - \hat{\boldsymbol{x}}(t)$. Compute

$$T^2(t) = \sum_{k=1}^{r} \frac{t_k(t)^2}{\lambda_k}, \qquad Q(t) = \|\boldsymbol{e}(t)\|^2.$$

(iv) **Control limits:** set T^2_{crit} using an F-distribution approximation (under assumptions) or empirical quantiles from baseline; set Q_{crit} using Jackson–Mudholkar approximation or empirical quantiles.

(v) **After an alarm:** diagnose with contribution plots (which variables drive T^2 or Q), inspect raw signals, check sensor health, and relate to process events (setpoints, maintenance). Decide whether it is a fault, a new operating regime (may require model update), or bad data. For drift/nonstationarity consider moving-window PCA or periodic recalibration.

Question 16 *(Using moving-window PCA for drift)*

[8 points] You suspect the process relationships are slowly changing over months (nonstationary behavior).

(a) What is the idea of moving-window PCA?

(b) What trends in (i) variance explained and (ii) loadings over time indicate drift?

(c) What are two choices for the window step s and what do they mean?

Solution.

(a) Fit PCA repeatedly on a window of the most recent data of length w (e.g., last w time points) to obtain time-varying PCs. Track eigenvalues/variance explained, loadings, and monitoring statistics as the window moves.

(b) Drift indicators: (i) explained variance for leading PCs changes (often drops if new variance appears outside the current model), and (ii) loadings rotate/shift, indicating changing correlations/relationships among variables.

(c) Step choices: **sliding/overlapping** ($s = 1$ or small) updates frequently with overlap; **block/non-overlapping** ($s = w$) fits on disjoint blocks for cheaper computation but slower detection.

Question 17 *(Interpreting an alarm: T^2 high vs Q high)*

[8 points] During monitoring you see three situations:

(a) T^2 crosses its limit while Q remains normal,

(b) Q crosses its limit while T^2 remains normal,

(c) both cross their limits.

For each, state the most likely interpretation and one diagnostic plot you would use next.

Solution.

(a) High T^2, normal Q: the sample is extreme within the learned model (unusual combination of normal patterns). Next: T^2 contribution plot (which variables/scores drive it), score plot with ellipse, context check (operating point change).

(b) High Q, normal T^2: new/unmodeled behavior (residual structure). Next: Q (SPE) contribution plot or residual heatmap to identify variables with large residuals; check sensor faults/nonlinearities; consider more PCs or model update.

(c) Both high: strong outlier/fault candidate. Next: both contribution plots and raw signal inspection; likely root-cause investigation and possible exclusion from model.